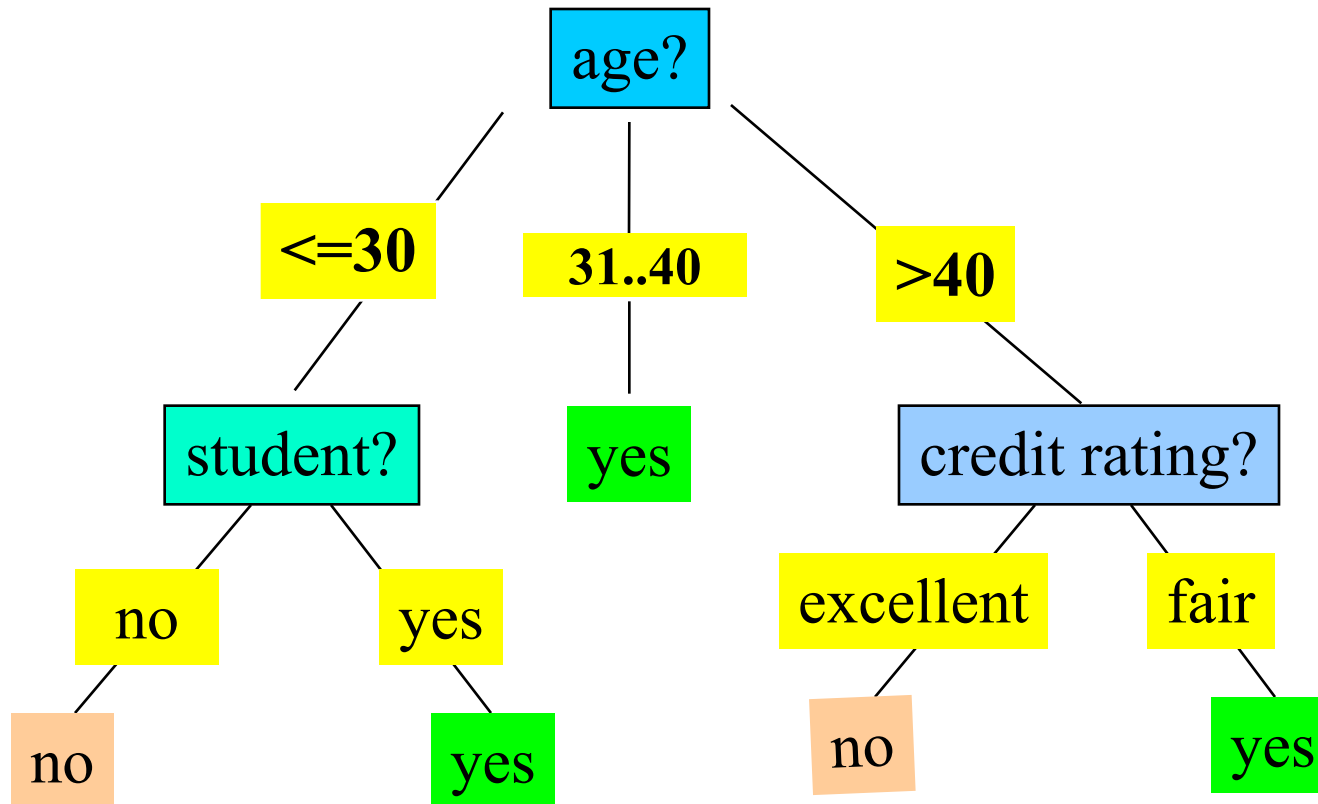


Decision Tree Induction: Training Dataset

This
follows an
example
of (Buy
Computer)

age	income	student	credit rating	buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

Output: A Decision Tree for “*buys_computer*”



Algorithm for Decision Tree Induction

❖ Basic algorithm (a greedy algorithm)

1. Tree is constructed in a **top-down recursive divide-and-conquer manner**
2. At start, all the training examples are at the root
3. **Attributes are categorical** (if continuous-valued, they are **discretized in advance**)
4. Examples are partitioned recursively based on selected attributes
5. Test attributes are selected **on the basis of a heuristic or statistical measure** (e.g., **information gain, gain ratio, gini index**)

❖ Conditions for **stopping partitioning**

- All samples for a given node belong to the same class
- There are no remaining attributes for further partitioning
 - **majority voting** is employed for classifying the leaf
- There are no samples left

Attribute Selection Measure: Information Gain (ID3/C4.5)

- Select the attribute with the **highest information gain**
- Let p_i be the probability that an arbitrary tuple in D belongs to class C_i , estimated by $|C_{i,D}|/|D|$
- **Expected information** (entropy) needed to classify a tuple in D :

$$Info(D) = -\sum_{i=1}^m p_i \log_2(p_i)$$

- **Information** needed (after using A to split D into v partitions) to classify D :

$$Info_A(D) = \sum_{j=1}^v \frac{|D_j|}{|D|} \times I(D_j)$$

- **Information gained** by branching on attribute A

$$Gain(A) = Info(D) - Info_A(D)$$

Attribute Selection: Information Gain

g Class P: buys_computer = "yes"

g Class N: buys_computer = "no"

$$Info(D) = I(9,5) = -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) = 0.940$$

age	p_i	n_i	$I(p_i, n_i)$
≤ 30	2	3	0.971
31...40	4	0	0
> 40	3	2	0.971

$$Info_{age}(D) = \frac{5}{14} I(2,3) + \frac{4}{14} I(4,0) + \frac{5}{14} I(3,2) = 0.694$$

$\frac{5}{14} I(2,3)$ means "age ≤ 30 " has 5 out of 14 samples, with 2 yes'es and 3 no's. Hence

$$Gain(age) = Info(D) - Info_{age}(D) = 0.246$$

Similarly,

$$Gain(income) = 0.029$$

$$Gain(student) = 0.151$$

$$Gain(credit_rating) = 0.048$$

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Attribute Selection: Information Gain

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14	>40	medium	no	excellent	no

$$C_1 = \text{Yes} = 9$$

$$C_2 = \text{No} = 5$$

1- compute $info(D)$

$$= - \sum_{i=1}^m p_i \log_2(p_i)$$

$$= - \frac{9}{14} \log_2 \left(\frac{9}{14} \right) - \frac{5}{14} \log_2 \left(\frac{5}{14} \right)$$

$$= 0.940 \text{ bits}$$

2- compute $info_{age}(D)$

$$= \sum_{j=1}^v \frac{|D_j|}{|D|} \times info(D_j)$$

$$= \frac{5}{14} \times \left(- \frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} \right)$$

$$+ \frac{4}{14} \times \left(- \frac{4}{4} \log_2 \frac{4}{4} \right)$$

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$$= 0.694 \text{ bits}$$

3- compute $Gain(age) = 0.94 - 0.694 = 0.246 \text{ bits}$

Attribute Selection: Information Gain

RID	age	income	student	Credit_rating	Class: buys_computer
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2	<=30	high	no	excellent	no
3	31...40	high	no	fair	yes
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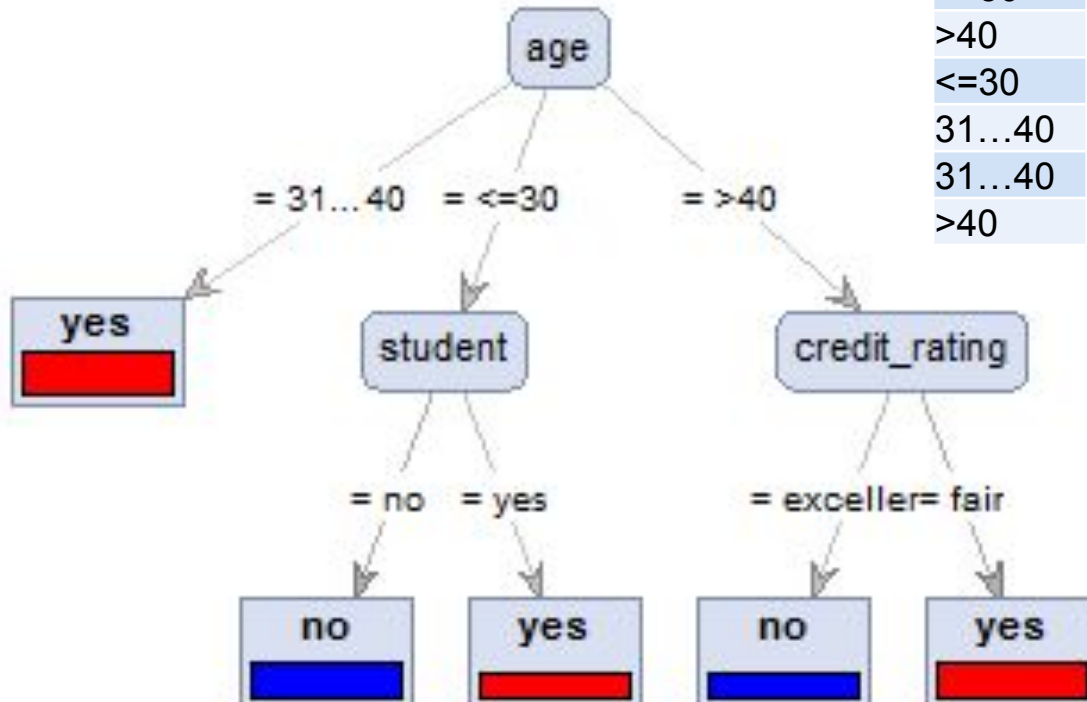
Gain(age) = 0.246 bits
Gain(income) = 0.029 bits
Gain(student) = 0.151 bits
Gain(credit_rating) = 0.048 bits

Gain(age) has highest information gain

Attribute Selection: Information Gain

❖ Training data set:
Buys_computer

age	income	student	credit rating	buys computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
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>40	low	yes	excellent	no
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<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no



Practice : Training Data

No	OUTLOOK	TEMPERATURE	HUMIDITY	WINDY	PLAY
1	Sunny	Hot	High	FALSE	No
2	Sunny	Hot	High	TRUE	No
3	Cloudy	Hot	High	FALSE	Yes
4	Rainy	Mild	High	FALSE	Yes
5	Rainy	Cool	Normal	FALSE	Yes
6	Rainy	Cool	Normal	TRUE	Yes
7	Cloudy	Cool	Normal	TRUE	Yes
8	Sunny	Mild	High	FALSE	No
9	Sunny	Cool	Normal	FALSE	Yes
10	Rainy	Mild	Normal	FALSE	Yes
11	Sunny	Mild	Normal	TRUE	Yes
12	Cloudy	Mild	High	TRUE	Yes
13	Cloudy	Hot	Normal	FALSE	Yes
14	Rainy	Mild	High	TRUE	No

Practice: Training Data (Entropy Calculations)

❖ Entropy **Total**

$$Entropy(Total) = (-\frac{4}{14} * \log_2(\frac{4}{14})) + (-\frac{10}{14} * \log_2(\frac{10}{14}))$$

$$Entropy(Total) = 0.863120569$$

❖ Entropy (**Outlook**)

$$Entropy(Cloudy) = (-\frac{0}{4} * \log_2(\frac{0}{4})) + (-\frac{4}{4} * \log_2(\frac{4}{4})) = 0.000000000$$

$$Entropy(Rainy) = (-\frac{1}{5} * \log_2(\frac{1}{5})) + (-\frac{4}{5} * \log_2(\frac{4}{5})) = 0.721928095$$

$$Entropy(Sunny) = (-\frac{3}{5} * \log_2(\frac{3}{5})) + (-\frac{2}{5} * \log_2(\frac{2}{5})) = 0.970950594$$

❖ Entropy (**Temperature**)

$$Entropy(Cool) = (-\frac{0}{4} * \log_2(\frac{0}{4})) + (-\frac{4}{4} * \log_2(\frac{4}{4})) = 0.000000000$$

$$Entropy(Hot) = (-\frac{2}{4} * \log_2(\frac{2}{4})) + (-\frac{2}{4} * \log_2(\frac{2}{4})) = 1.000000000$$

$$Entropy(Mild) = (-\frac{2}{6} * \log_2(\frac{2}{6})) + (-\frac{4}{6} * \log_2(\frac{4}{6})) = 0.918295834$$

❖ Entropy (**Humidity**)

$$Entropy(High) = (-\frac{4}{7} * \log_2(\frac{4}{7})) + (-\frac{3}{7} * \log_2(\frac{3}{7})) = 0.985228136$$

$$Entropy(Normal) = (-\frac{0}{7} * \log_2(\frac{0}{7})) + (-\frac{7}{7} * \log_2(\frac{7}{7})) = 0.000000000$$

❖ Entropy (**Windy**)

$$Entropy(False) = (-\frac{2}{8} * \log_2(\frac{2}{8})) + (-\frac{6}{8} * \log_2(\frac{6}{8})) = 0.811278124$$

$$Entropy(True) = (-\frac{4}{6} * \log_2(\frac{4}{6})) + (-\frac{2}{6} * \log_2(\frac{2}{6})) = 0.918295834$$